Decoding Learning Behaviors: Interpretable Profiles and Their Impact on Academic Outcomes

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ABSTRACT

For a long time, educators have been concerned with how to learn effectively. With the growing popularity of online learning platforms, we aim to study user behavioral data from an educational platform, GoGymi, in order to identify behavioral patterns that are beneficial to learning outcomes. We extracted meaningful features from the behavioral data and grouped them into four major learning dimensions. Based on these dimensions, we constructed Learning Profiles to characterize each student's behavioral type. Beyond clustering, we also sought to explain the importance of different dimensions and features for learning outcomes. To this end, we employed both linear and non-linear models for estimation and used interpretability tools to help us understand the results. Our findings suggest that practicing previously unseen questions, watching instructional videos to completion, increasing interactions with the chatbot, and spending more time on learning all have significant positive effects on academic performance.

1. INTRODUCTION

With the growing prevalence of online educational platforms, researchers are increasingly able to examine the relationship between learning processes and academic performance through students' behavioral traces in digital learning environments. Compared with relying solely on exam scores or questionnaire data, platform-generated behavioral data provide a richer account of students' actual learning activities, such as practice behavior, content engagement, resource use, and time investment [7]. This creates new opportunities to better understand how students learn and how such behaviors relate to academic outcomes.

Prior research has suggested that behavioral trace data can capture students' learning processes more directly than self-reported measures [22]. For example, Zhou and Winne found that traced goal orientations outperformed self-reported goals in predicting academic achievement, indicating that indicators derived from platform behaviors may not only complement the limitations of self-reported data, but also more effectively capture behavior patterns associated with learning outcomes [22]. Such findings highlight both the methodological feasibility and the educational value of studying student performance through behavioral data collected from online learning platforms.

In the field of learning analytics, a substantial body of work has shown that student behavioral data can be used to predict academic performance and support tasks such as early warning and targeted intervention [10, 21]. However, much of this research has primarily focused on improving predictive performance, with relatively less attention paid to identifying which learning behavior dimensions matter most and how these relationships can be interpreted in an educationally meaningful way. In other words, while existing studies have established that behavioral data are useful for predicting performance, there remains a need to better understand which behaviors deserve particular attention and how their influence on academic outcomes can be explained interpretably.

Meanwhile, student profiling, although not an entirely new research direction, has also been explored in prior studies [19, 16]. Existing work has attempted to group students according to their learning behaviors in order to identify different learning patterns or determine which students may require additional learning support [19, 16]. Nevertheless, such studies are often conducted within specific courses or relatively constrained learning settings, leaving open the question of whether these behavioral profiles remain meaningful in larger-scale and more authentic online learning contexts. In comparison, the GoGymi dataset used in this study includes 531 students, providing an opportunity to re-examine the relationship between behavioral profiles and academic performance in a larger sample and in a realistic exam-preparation setting.

Against this background, the present study investigates the relationship between students' learning behavior patterns and academic performance using behavioral data collected from 531 students on the GoGymi platform. Specifically, meaningful behavioral features are first extracted from the platform data and organized into four learning dimensions, which are then used to construct interpretable Learning Profiles. In addition, both linear and non-linear models are employed to analyze the effects of different behavioral dimensions and features on academic performance. Interpretabil-

ity tools are further applied to provide a more systematic explanation of which learning behaviors are most strongly associated with better academic outcomes.

This study addresses the following two research questions:

- **RQ1:** Can we identify interpretable behavioral profiles of students that are meaningfully related to academic performance?
- **RQ2:** Which behavioral features and dimensions most strongly drive academic performance, and how can these relationships be explained interpretably?

These two research questions correspond to two complementary levels of analysis. The first focuses on the group level, examining whether students can be meaningfully characterized through interpretable behavioral profiles. The second focuses on the feature level, investigating which specific behavioral dimensions and features are most strongly associated with academic performance. By integrating these two perspectives, this study aims not only to characterize the behavioral patterns of high-performing students, but also to explain more systematically which learning behaviors may contribute to better academic outcomes, thereby providing implications for future instructional support and personalized learning design.

2. RELATED WORK

Behavioral trace data and learning outcomes. Platform generated behavioral logs have been shown to predict academic performance more reliably than self-reported measures: Zhou and Winne found that traced goal orientations outperformed self-reported ones in predicting achievement [22], and Van Halem et al. confirmed that trace data explain a unique share of outcome variance beyond dispositional surveys, while also noting that student motivation remains partially covert and not fully captured by logs alone [20]. At the feature level, video engagement and practice breadth have emerged as consistent positive signals across platforms: completion, pausing, and replay behavior correlate with better learning outcomes, while high-achieving students tend to explore more resources and use help-seeking tools more actively.

Student profiling via clustering. Tempelaar et al. used cluster analysis on e-tutorial trace data to construct learning profiles characterized by students' dispositions, finding trace-based profiling more actionable than self-reports for driving educational interventions [19]. Seo et al. extended this toward a prescriptive framing, deriving profiles from observational data to support scalable, tailored student support [16]. Both studies, however, cluster all features jointly, conflating heterogeneous behaviors into a single space. Mejia-Domenzain et al. addressed this by clustering students separately per behavioral dimension before integrating the labels into composite profiles, preserving end-to-end interpretability [12, 11]. Our pipeline follows this dimension-first design and applies it to a larger, real-world exam-preparation context.

Predicting performance and explaining it. A broad literature confirms that behavioral features carry predictive signal for academic outcomes, with the primary motivation being early warning systems [10, 21]. Yet, coarse summary statistics have limited capacity to reveal the strategies students actually adopt, calling for approaches that go beyond count-based predictors. We respond by pairing regression with TreeSHAP attribution [8], which exposes directional contrasts invisible to accuracy metrics. Because importance rankings can be an artefact of the chosen explainer [18], we cross-examine SHAP against LIME across all 531 students—a full-cohort implementation of the evaluation protocol proposed by Swamy et al. [18].

3. METHODOLOGY

Dimension	Feature	Description
Effort	Active days	Number of distinct days with platform activity
	Active learning time	Total time online, idle periods excluded
Content Depth	Avg. video completion	Mean fraction of each video watched
	Video completion rate	Share of videos watched to $\geq$90%
	Videos interacted	Total number of videos engaged with
	Pauses per video	Mean pause events per video
	Replays per video	Mean replay events per video
	Avg. scroll depth	Mean fraction of page scrolled
Help Seeking	Chatbot messages	Mean user messages per chatbot thread
	Prompt effort	Mean character length of user prompts
Practice Review	Repeated question views	Questions visited more than once
	Unique questions viewed	Distinct questions navigated to
	Total question views	All question navigation events

Table 1: Behavioural features used for student clustering.

Our two research questions operate at different levels of analysis and therefore call for different modelling strategies. RQ1 is an unsupervised, group-level question, while RQ2 is a supervised, feature-level question. The first subsection describes how raw platform events are cleaned, imputed, and featurised into the per-student feature table that both questions draw on. The second subsection describes the models chosen for each question and the rationale behind them, and the third details the evaluation metrics used to validate each stage and why they are appropriate.

3.1 Data Preprocessing and Featurization

Raw events are transformed into a flat, per-student feature table. The thirteen resulting features are grouped into four conceptual dimensions: Effort, Content Depth, Help Seeking, and Practice Review, plus one outcome variable held out for validation. These dimensions are grounded in prior behavioral trace research, which consistently identifies time investment, content engagement, help-seeking, and practice as the core axes along which student self-regulation manifests in platform logs [12, 14, 2]. Table 3 provides a full

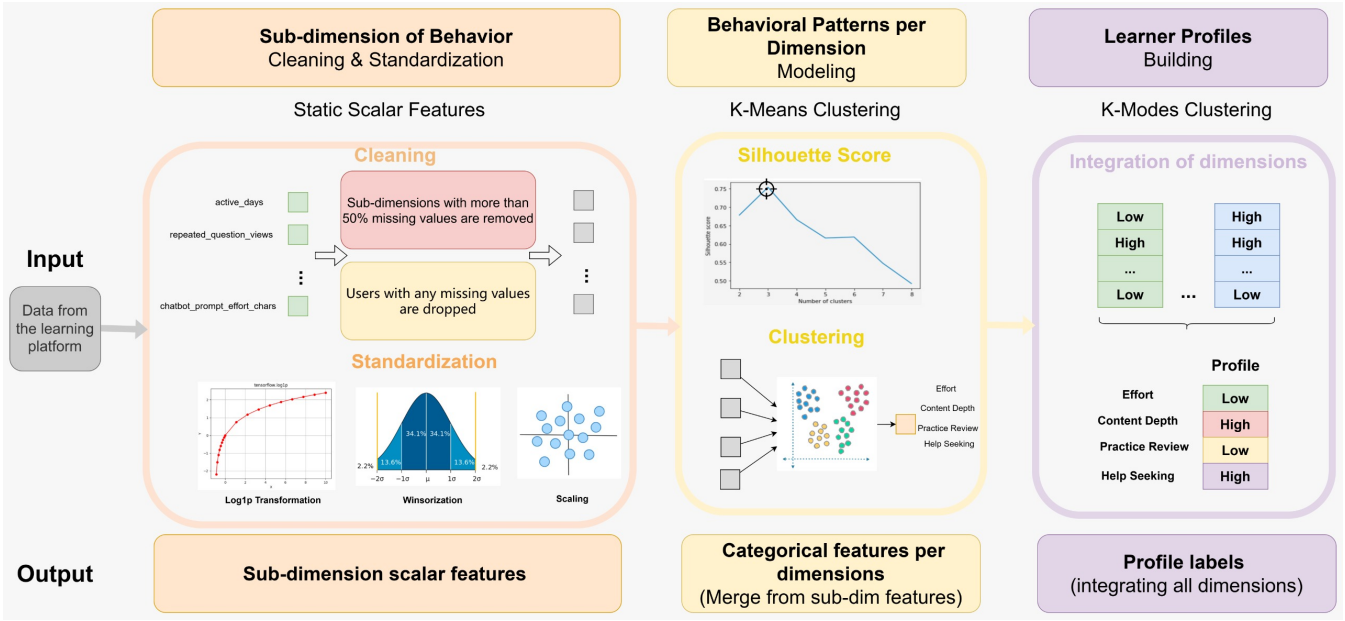


Figure 1: Architecture for Building Student Profiles: Platform data is cleaned and standardized into features, each categorized into a corresponding dimension. K-Means clustering is applied to each dimension, assigning every student a cluster label per dimension. These labels are combined to form student profiles, which are then clustered using K-Modes to generate the final learner profiles.

inventory of features. Finally, after cleaning and preprocessing, we have a dataset of 531 students left out of 1781 students.

Outcome Variable. Quiz and math records are pooled; each attempt yields a score ratio of `points/max_points`. The student-level mean, `average_score_ratio`, is excluded from all clustering steps of RQ1 and used only at the results stage to prevent any leakage.

Scaling. Dimension-specific scaling is applied:

- **Effort:** features have gaussian shape hence, they are standardised with `StandardScaler` (zero mean, unit variance).
- **Content Depth, Help Seeking, Practice Review:** count-based features are right skewed, hence they are first transformed with $\log(1+x)$, then winsorised at the 1st and 99th percentiles to remove outliers. `RobustScaler` (centres on median, scales by IQR) is applied last, providing further resistance to remaining extremes.

3.2 Models per Research Question

3.2.1 *RQ1: Identifying Interpretable Learning Profiles*

To characterise students through interpretable behavioural profiles, we build on the multi-step clustering pipeline of Mejia-Domenzain et al. [12, 11], whose central design principle is to cluster learners *separately* along theoretically grounded behavioural dimensions and only afterwards integrate those dimension-level patterns into multidimensional profiles. We follow this principle because the per-dimension decomposition is what makes the profiles interpretable: clustering in a single joint feature space would mix incommensurable behaviours (e.g. video pauses with chatbot turns) and lead to uninterpretable centroids, whereas clustering each of our four dimensions (Effort, Content Depth, Practice Review, and Help Seeking) independently means every resulting label retains a clear behavioural meaning.

Within each dimension, we use **K-Means**. K-Means is well suited here because the features within a dimension are continuous and standardised, and its hard assignments yield the interpretable discrete labels that the subsequent profiling stage requires. The number of clusters K is selected per dimension by grid search over $\{2, \ldots, 8\}$ with 20 random restarts per candidate and a fixed seed, retaining the partition that maximises the silhouette score.

The four dimension labels are then assembled into a categorical matrix - one row per student, one column per dimension. The students are then clustered with **K-Modes** to discover composite learner profiles capturing co-occurring behavioural patterns. K-Modes extends K-Means to categorical data by replacing Euclidean distance with Hamming distance and representing cluster centres as modes rather than means, making it the natural choice for our purely categorical label matrix. We use the Huang initialisation [5] since it seeds modes from attribute-value frequencies rather than at random, producing more representative starting points and reducing sensitivity to poor initial centroids. The number of clusters K is selected by grid search over $\{2, \ldots, 8\}$ with 10 random restarts per K, retaining the partition that maximises the hamming-based silhouette score. This two-stage K-Means/K-Modes design keeps each modelling step

matched to its data type (continuous then categorical) while preserving end-to-end interpretability.

3.2.2 RQ2: Quantifying and Explaining Behavioural Drivers

RQ2 asks which dimensions and features most strongly drive performance, so we predict the continuous outcome `average_score_ratio` using two complementary model families: linear models for direct, coefficient-based interpretability, and non-linear ensembles paired with post-hoc explainers for capturing interactions the linear models cannot.

Linear models. We fit a two-level hierarchical Ordinary Least Squares regression. The first level regresses performance on all four discrete dimension labels simultaneously, so each coefficient gives that dimension's direction and relative importance while controlling for the others. The second level regresses performance on the raw features within each dimension, isolating each feature's independent contribution. Linear regression is chosen for this stage precisely because its coefficients are directly interpretable as marginal effects, which answers the "which and in what direction" part of RQ2 transparently. Before fitting, we guard against multicollinearity using the Variance Inflation Factor (VIF) alongside pairwise correlation checks. This resulted in the removal of three features—active days, average video completion, and repeated question views—as each was found to be collinear with a corresponding retained feature: active learning time, video completion rate, and total question views, respectively.

Non-linear models. Because linear regression cannot capture feature interactions, we complement it with two tree ensembles. **XGBoost** [4] is our primary non-linear model: it is a strong gradient-boosted tree learner and, crucially, admits exact and efficient Shapley-value attribution via TreeSHAP [8]. A **Random Forest** [3] is trained as a robustness check; agreement between the two models' importance rankings makes our conclusions substantially more credible by showing that importance rankings are not an artefact of a single learner. Both are fit as regressors on the continuous `average_score_ratio` using the same VIF-screened set of ten features used for the linear models — the thirteen behavioural features minus the three dropped for multicollinearity — keeping the linear and non-linear analyses directly comparable.

Post-hoc explainability. To open up the ensembles, we use **SHAP** [9] as the primary explanation method, computing exact attributions with TreeSHAP [8]. SHAP decomposes each prediction into additive per-feature contributions; we aggregate the mean absolute attributions into a global feature-importance ranking, sum them within each dimension to obtain a dimension-level ranking, and read off the direction of each effect via beeswarm plots, supplemented by the Pearson correlation between each feature's raw value and its SHAP attribution across all students — a positive correlation indicates the feature raises scores when high, a negative one that it lowers them. Because the choice of explainer can itself drive the resulting feature-importance scores [18], we do not rely on SHAP alone but cross-examine it against **LIME** [13] applied to the same model. Following Swamy et al. [18], we compute each method's local importance vector for *every* student and quantify their agreement with both a rank-based measure (Spearman's rank-order correlation) and a distribution-based measure (Jensen-Shannon distance), rather than inspecting a single representative case. We favour SHAP as our main tool because it offers theoretical guarantees (local accuracy, missingness, consistency), supports both local and global analysis, and is deterministic, whereas LIME is local-only and provides run-to-run variable explanations; LIME therefore serves as an independent corroborating explainer rather than the primary one.

3.3 Evaluation Metrics

Cluster validity (RQ1). Because clustering is unsupervised, we select K and assess partition quality with the **silhouette coefficient** [15], which measures how well each point fits its own cluster relative to the nearest alternative and requires no ground-truth labels. We favour the silhouette over other model-selection heuristics (e.g. the elbow method) because it is bounded in $[-1, 1]$ and easy to interpret. For K-Means we use the standard Euclidean silhouette; for the categorical K-Modes stage we use a Hamming-based silhouette, giving a normalised quality measure that is directly comparable across both clustering stages.

Stability of the K-Means partitions (RQ1). We do not rely on the silhouette alone but add two complementary checks on the 531 profiled students. A bootstrap stability check refits K-Means on 100 random 80% subsamples and compares each partition to the full-cohort labels via the Adjusted Rand Index, testing whether the discrete labels are reproducible under resampling rather than an artefact of a single fit. A random two-way labelling baseline tests whether the K-Means silhouette exceeds what an arbitrary split would achieve; we run 20 independent initialisations per fit so the reported evaluation is robust to initialisation. These checks, together with the held-out validation below, guard against mistaking spurious structure for genuine behavioural groups.

Profile validation (RQ1). The outcome variable is held out of all clustering and used only to test whether the discovered profiles differ in academic performance, in order to establish that the profiles are educationally meaningful. A **Shapiro–Wilk** test [17] first shows that scores are not normally distributed ($p < 0.05$), so we adopt the **Kruskal–Wallis** H test [6], a non-parametric alternative to one-way ANOVA that is robust to non-normality and to the unequal group sizes across profiles — to assess whether score differences across profiles exceed chance, at a threshold of $p < 0.05$.

Linear model evaluation (RQ2). For the hierarchical regression, we report the model-level R^2 and adjusted R^2 together with the overall F-test, and judge individual predictors by their coefficient t-statistics, p-values, and 95% confidence intervals; we additionally report a 5-fold cross-validated R^2 at both the dimension and feature levels to gauge out-of-sample generalisation rather than relying on in-sample fit alone. Before fitting, we screen for **multicollinearity** using the Variance Inflation Factor (iterative removal at threshold VIF < 5) and inspect pairwise correlations; this is essential because collinear behavioural features (e.g. active days and total active time) would otherwise destabilise and mislead the coefficient interpretation that the linear stage depends

on.

Non-linear model evaluation (RQ2). The ensembles are evaluated with 5-fold cross-validation, reporting the cross-validated R^2 on the continuous outcome. We report cross-validated rather than in-sample scores to guard against overfitting. Predictive accuracy here is secondary to explanation: the metrics check if the models capture genuine signal, while the interpretability of the SHAP attributions is what ultimately answers RQ2. Accordingly, the credibility of those attributions is assessed on two fronts. First, across models, by the **Spearman rank correlation** between the global SHAP importance rankings of XGBoost and Random Forest. Second, across explainers, by cross-examining SHAP against LIME on the XGBoost model following the protocol of Swamy et al. [18]: for every one of the 531 students, we compare the two methods' local importance vectors using both **Spearman's rank-order correlation** (agreement on feature ranking) and the **Jensen–Shannon distance** (agreement on the normalised importance values), and report the resulting distributions over the full cohort. Because our tree-model explainers are inexpensive, we evaluate the entire cohort instead of the sampled representative subset used by Swamy et al. [18], whose counterfactual explainers made full-cohort analysis computationally infeasible.

4. RESULTS

4.1 RQ1: Interpretable Learning Profiles

Per-dimension structure. The silhouette grid search selects $K = 2$ for all four dimensions, yielding two opposed, easily named groups. Separation is strong for *Effort* (0.54) and *Practice Review* (0.53) and more modest for *Help Seeking* (0.38) and *Content Depth* (0.26); the lower *Content Depth* value reflects the six correlated, heavy-tailed engagement features it pools rather than an absence of structure. The splits are interpretable and informatively unbalanced: (152 *High Effort*/379 *Low Effort*), (382 *Deep Engagers*/ 149 *Surface Browsers*), (423 *Deep Reviewers*/108 *Light Reviewers*), (147 *Deliberate Askers*/384 *Passive Chatters*), and the centroids confirm the labels: *Deep Engagers* watch 74% of the average video versus 37% for *Surface Browsers*, and *Deep Reviewers* revisit roughly seven times as many questions as *Light Reviewers*.

Validation: are the splits real and reproducible? Two checks (Figure 2) confirm the structure is not an artefact of the silhouette. A random two-way relabelling collapses the silhouette to ≈ 0 in every dimension, and a bootstrap over 100 random 80% subsamples reproduces the full-cohort labels with high agreement (Adjusted Rand Index ≥ 0.90 throughout), so the binary labels are reproducible under resampling rather than a single-fit accident. The per-dimension labels are therefore a stable categorical input to the next stage of K-Modes.

Composite profiles. K-Modes over the four-label matrix recovers eight composite profiles. The Hamming silhouette rises with K; we retain $K = 8$, the maximum-silhouette solution (0.81), as it stays interpretable with every profile well populated (≥ 31 students). Table 2 summarises the profiles and their dimension modes.

Profiles relate to performance. With `average_score_ratio` held out of clustering, we validate the profiles without leakage. Scores are non-normal (Shapiro–Wilk $W = 0.97$, $p < 10^{-8}$), so a Kruskal–Wallis test confirms the profiles differ significantly in performance ($H = 18.96$, $p = 8.3\times10^{-3}$; Figure 3). The ranking is coherent: the top four profiles (325 students, 61% of the cohort) are *uniformly* Deep Engagers and Deep Reviewers, and no profile with Surface Browsing or Light Reviewing reaches the upper half. Crucially, *targeted* engagement beats raw time — the Low-Effort Profile 5 ranks #2, above the High-Effort Profile 2 at #3. Also, with the other dimensions fixed, *Deliberate Askers* outscore *Passive Chatters* (Profile 4 > Profile 2; Profile 5 > Profile 7).

Additional finding: do top profiles learn more consistently? Beyond mean level, the upper-half profiles also show lower within-profile score dispersion than the lower half (0.144 vs. 0.181 after centering each student on their profile median); a Fligner–Killeen test (homogeneity-of-variance test which is robust to non-normality) supports the difference (one-sided

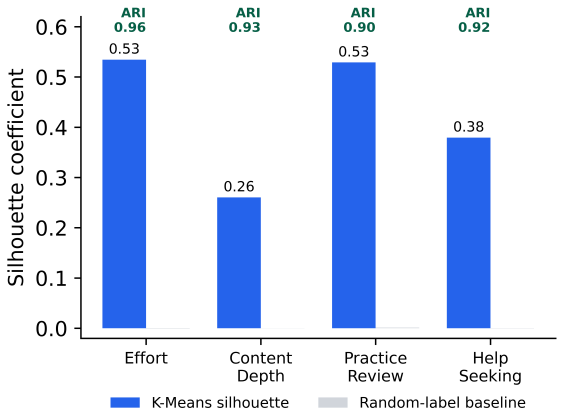


Figure 2: Validation of the RQ1 per-dimension clustering: K-Means silhouette versus a random-label baseline (≈ 0), with the bootstrap Adjusted Rand Index (ARI ≥ 0.90) annotated above each bar, showing the splits are both genuine and reproducible.

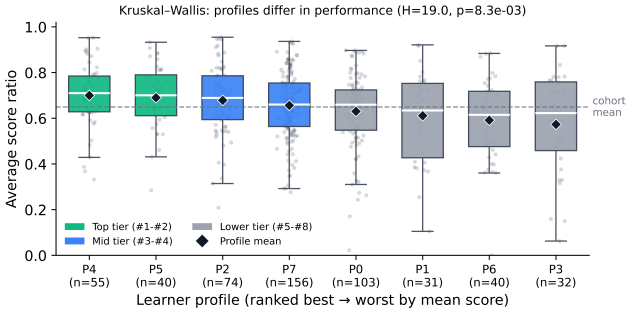


Figure 3: Distribution of held-out academic performance (`average_score_ratio`) across the eight learner profiles, ranked best to worst by mean. Boxes are coloured by performance tier; diamonds mark profile means; the dashed line is the cohort mean. A Kruskal–Wallis test confirms the profiles differ significantly.

Table 2: Profile score summary, sorted by mean `average_score_ratio` (best first). Coloured dimension labels mark the behaviourally "positive" trait of each dimension; grey marks the opposing trait. Score is the held-out academic outcome, never used during clustering.

Profile	Students	Mean ± Std Score	Rank	Effort	Content Depth	Practice Review	Help Seeking
4	55	0.700±0.14	#1	High Effort	Deep Engagers	Deep Reviewers	Deliberate Askers
5	40	0.690±0.14	#2	Low Effort	Deep Engagers	Deep Reviewers	Deliberate Askers
2	74	0.678±0.15	#3	High Effort	Deep Engagers	Deep Reviewers	Passive Chatters
7	156	0.656±0.14	#4	Low Effort	Deep Engagers	Deep Reviewers	Passive Chatters
0	103	0.630±0.16	#5	Low Effort	Surface Browsers	Deep Reviewers	Passive Chatters
1	31	0.611±0.20	#6	Low Effort	Deep Engagers	Light Reviewers	Deliberate Askers
6	40	0.591±0.18	#7	Low Effort	Deep Engagers	Light Reviewers	Passive Chatters
3	32	0.573±0.22	#8	Low Effort	Surface Browsers	Light Reviewers	Passive Chatters

$p = 6.0 \times 10^{-3}$). High-performing students thus yield not only higher but more reliable attainment.

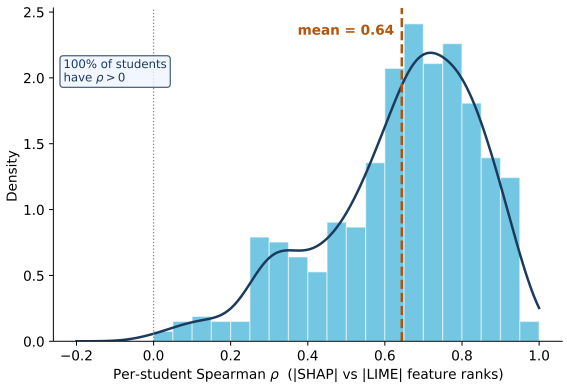


Figure 4: Per-student rank agreement between absolute SHAP and LIME feature importances on the XGBoost model across all 531 students is positive for ≥ 99% of students

4.2 RQ2: Behavioural Drivers of Performance

Linear models: which dimensions and features matter. The Level-1 hierarchical OLS, regressing performance on the four binary dimension labels simultaneously, finds *Content Depth* ($\beta = 0.036$, $p = 0.020$) and *Practice Review* ($\beta = 0.056$, $p = 0.002$) significantly positive while controlling for the others; *Effort* ($p = 0.10$) and *Help Seeking* ($p = 0.42$) are not significant at the dimension level. The model is jointly significant (F-test $p = 5.5 \times 10^{-5}$) though it explains little variance (in-sample $R^2 = 0.046$), foreshadowing that behaviour captures only a modest slice of attainment. Descending to features (Level 2), the significant positive drivers are *active learning time* (Effort, $p < 10^{-3}$), *video completion rate* (Content Depth, $p = 0.002$), *unique questions viewed* (Practice Review, $p = 0.005$) and *chatbot messages* (Help Seeking, $p = 0.045$).

Overfitting check: in-sample fit overstates generalisation. XGBoost reaches $R^2 = 0.33$ in-sample but only 0.05 ± 0.03 under 5-fold cross-validation, with the Random Forest having identical CV score (0.05±0.04). Engagement behaviour thus carries a small but real signal — enough to rank drivers, not to predict performance scores.

Non-linear models: importance and direction. TreeSHAP on XGBoost ranks the primary drivers of each dimension: *active learning time* > *video completion rate* > *chatbot messages per thread* > *unique questions viewed* (Figure 5). Within Content Depth, *completing* videos raises scores (corr = +0.83) whereas touching many videos, replaying them, and scrolling more deeply all carry negative directions (corr = −0.47, −0.29, −0.32 respectively) — depth of viewing matters, not churn; pauses per video show negligible directional effect (corr ≈ −0.04). Within Practice Review, *unique* questions viewed is strongly positive (corr = +0.89) while total question views is only weakly positive (corr = +0.11) — breadth of exposure matters more than raw visit volume. Sustained chatbot dialogue (messages per thread) is positive (corr = +0.61), echoing the Deliberate-Asker advantage from RQ1. Notably, within Help Seeking, messages per thread outranks prompt effort in characters (corr = +0.29), indicating that iterative back-and-forth engagement predicts performance more than the elaborateness of any single message — it is the sustained exchange, not the well-crafted prompt, that matters.

Robustness check: are the explanations trustworthy? Because feature-importance scores can be an artefact of the chosen model or explainer [18], we corroborate them on two axes. Across models, the XGBoost and Random-Forest SHAP rankings agree almost perfectly (Spearman $\rho = 0.98$, $p = 10^{-6}$). Across explainers, we compare SHAP against LIME on the same XGBoost model following Swamy et al. [18], computing both methods' local importance vectors for *every* one of the 531 students: per-student rank agreement is positive for ≥ 99% of students (mean Spearman $\rho \approx 0.64$) (Figure 4) and the normalised-value distance is small (mean Jensen–Shannon ≈ 0.27, ≥ 99% below 0.5) (Figure 6). The leading drivers are thus stable to both the learner and the explainer, and — reassuringly — *they coincide with the significant coefficients from the independent hierarchical linear analysis.*

5. DISCUSSION

The key to high-performing students lies not in spending more time, but in engaging with and reviewing learning ma-



Figure 5: SHAP analysis of the XGBoost regressor, features grouped by dimension. Left: global importance (mean $|\phi|$ in score-ratio units), bars coloured by dimension. Right: per-student SHAP beeswarm with dot colour encoding the feature value (low → high); the number beside each row is the correlation between the feature's value and its SHAP value — the direction of its effect (green +, red −, grey near-zero).

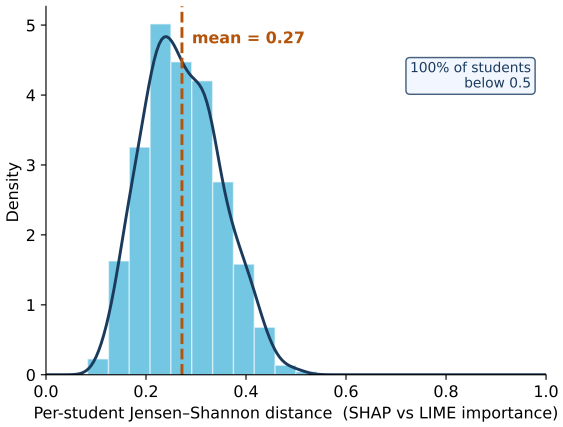


Figure 6: Distribution of per-student Jensen–Shannon distance between normalised SHAP and LIME feature-importance vectors on the XGBoost model ($n = 531$; $0 =$ identical distributions, $1 =$ maximally different). The mean distance is ≈ 0.27 and $\geq 99\%$ of students fall below 0.5, indicating strong agreement between the two explainers across the full cohort

terials in a higher-quality way. We found that the top four learner profiles in the study almost all shared two characteristics: Deep Engagers and Deep Reviewers. In contrast, any profile marked by Surface Browsing or Light Reviewing failed to rank in the upper half of performance. These four classes are derived from the dimensions of Content Depth and Practice Review, both of which emphasize the quality rather than the duration of learning. This suggests that learning quality plays a more important role in the learning process. More importantly, we found that one Low-Effort profile ranked even higher than a High-Effort profile, indicating that effort level alone is not sufficient to explain academic performance. What truly matters is whether students engage deeply with the content, actively revisit and review problems, and use learning resources in a strategic way. This finding runs counter to the traditional educational assumption that spending more time studying necessarily leads to better learning outcomes [1]. If platforms or instructors rely only on time-on-task, active minutes, or login frequency as indicators of learning engagement, they may overestimate superficial activity while underestimating genuinely effective learning behaviors. Implementing features like the automated retrieval of past concepts to facilitate deep review sessions, review cycles, embedding interactive checkpoints within videos and readings to discourage premature drop-offs, etc. will reliably help the students.

What contributes most to academic performance is not the sheer quantity of activity, but the pursuit of both depth and breadth of knowledge. In both the linear and non-linear analyses, the stable positive factors included video completion rate, unique questions viewed, and sustained chatbot dialogue. When students skim through videos superficially, they often miss the instructor's deeper explanations of the material. In many cases, course content is structured progressively, and deeper understanding depends on the buildup established in the earlier parts of a lesson. If students skip ahead after watching only the easier introductory content,

they may miss the more substantial concepts presented later, causing their learning to remain shallow. In addition, actively attempting a wider variety of questions helps broaden one's knowledge base. Once students become familiar with more types of problems, they are more likely to encounter questions they have seen before in the future; even when facing unfamiliar ones, they will already have learned a wider range of problem-solving strategies. This is more effective than repeatedly practicing the same fixed set of questions, and it reflects the development of breadth. Finally, when interacting with AI, students should not merely ask for an answer. Instead, they should engage in multi-turn dialogue and clarify every detail they do not understand, which reflects the pursuit of depth. Overall, the features that proved beneficial for learning in our study all point toward the pursuit of greater depth and breadth of knowledge. If educational platforms directly equate the volume of clickstream activity with engagement quality, they may reach misleading conclusions. In terms of design, the focus should shift from frequency-based metrics to completion, coverage, and interaction continuity; otherwise, low-efficiency but high-frequency behaviors may easily be misclassified as effective learning.

6. LIMITATIONS

First, by aggregating raw event logs into a flat per-student feature table, this study loses much of the temporal structure of learning behavior, including sequencing and pacing over time. Future work could adopt longitudinal or sequential modeling, such as weekly or session-level features, to capture behavioral changes and profile transitions more directly.

Second, the current profiling pipeline clusters each dimension separately and then combines binary labels into final profiles, which improves interpretability but may oversimplify student heterogeneity. Future research could explore soft clustering, mixture models, or latent profile analysis so that students can belong to multiple profiles with different probabilities.

7. CONCLUSION

This study identifies several behavioral factors associated with online learning performance and aims to provide empirical insights for learning platform design, instructional support, and students' learning strategies. In particular, amid the rapid development of AI-assisted education, an important question that still deserves deeper investigation is how to design human–AI interactions that can genuinely enhance learning outcomes. It is important to note that this study only provides preliminary evidence of a positive association between higher interaction frequency and better learning performance, and does not support strong causal claims. Therefore, future research should combine more rigorous controlled experiments with more fine-grained educational theory analysis to further understand the mechanisms of interaction between AI and students in educational settings, as well as how these interactions influence learning outcomes.

8. SUGGESTIONS TO STARTUP

One feature we can capture in the AI chatbot is whether the student's issue was successfully resolved. This would enable more effective analysis of chatbot interactions by identifying the types of conversations are most likely to lead to successful resolution and completion.

If an AI chatbot could proactively ask follow-up questions, that would be a valuable improvement. In our observations, we found that multi-turn dialogue actually helps improve students' academic performance. One possible reason is that, through repeated questioning, students are able to gradually learn the concepts they do not fully understand. However, we cannot assume that every student is capable of clearly identifying what they do not understand. In such cases, AI support becomes important: it can proactively ask about key concepts in a problem and guide students step by step toward understanding, rather than simply providing the answer all at once.

9. REFERENCES

[1] F. A. Adebayo. Time management and students academic performance in higher institutions, nigeria a case study of ekiti state. *International Research in Education*, 3(2):1–12, 2015.

[2] N.-J. Akpinar, A. Ramdas, and U. Acar. Analyzing student strategies in blended courses using clickstream data. In *Proceedings of the 13th International Conference on Educational Data Mining*, 2020.

[3] L. Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.

[4] T. Chen and C. Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.

[5] Z. Huang. Extensions to the k-means algorithm for clustering large data sets with categorical values. *Data Mining and Knowledge Discovery*, 2(3):283–304, 1998.

[6] W. H. Kruskal and W. A. Wallis. Use of ranks in one-criterion variance analysis. *Journal of the American Statistical Association*, 47(260):583–621, 1952.

[7] C. Lang, A. F. Wise, A. Merceron, D. Gašević, and G. Siemens. What is learning analytics. *The handbook of learning analytics*, 2:8–18, 2022.

[8] S. M. Lundberg, G. Erion, H. Chen, A. DeGrave, J. M. Prutkin, B. Nair, R. Katz, J. Himmelfarb, N. Bansal, and S.-I. Lee. From local explanations to global understanding with explainable ai for trees. *Nature Machine Intelligence*, 2(1):56–67, 2020.

[9] S. M. Lundberg and S.-I. Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, volume 30, pages 4765–4774, 2017.

[10] B. Maraza-Quispe, E. Damian Valderrama-Chauca, L. Henry Cari-Mogrovejo, and J. Milton Apaza-Huanca. Predictive model of student academic performance from lms data based on learning analytics. In *Proceedings of the 13th International Conference on Education Technology and Computers*, pages 13–19, 2021.

[11] P. Mejia-Domenzain, M. Marras, C. Giang, A. Cattaneo, and T. Käser. Evolutionary clustering of apprentices' self-regulated learning behavior in learning journals. *IEEE Transactions on Learning*

Technologies, 15(5):579–593, 2022.

[12] P. Mejia-Domenzain, M. Marras, C. Giang, and T. Käser. Identifying and comparing multi-dimensional student profiles across flipped classrooms. In *International Conference on Artificial Intelligence in Education*, pages 90–102. Springer, 2022.

[13] M. T. Ribeiro, S. Singh, and C. Guestrin. "why should i trust you?" explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 1135–1144, 2016.

[14] I. Roll, R. S. J. d. Baker, V. Aleven, and K. R. Koedinger. Help helps, but only so much: Research on help seeking with intelligent tutoring systems. *International Journal of Artificial Intelligence in Education*, 24(3):292–311, 2014.

[15] P. J. Rousseeuw. Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20:53–65, 1987.

[16] E.-Y. Seo, J. Yang, J.-E. Lee, and G. So. Prescriptive analytics for student success in an online university: Drawing learning profiles from trace observations for tailored support. *Computers & Education*, 237:105384, 2025.

[17] S. S. Shapiro and M. B. Wilk. An analysis of variance test for normality (complete samples). *Biometrika*, 52(3-4):591–611, 1965.

[18] V. Swamy, B. Radmehr, N. Krco, M. Marras, and T. KÃ¤ser. Evaluating the explainers: Black-box explainable machine learning for student success prediction in MOOCs. In A. Mitrovic and N. Bosch, editors, *Proceedings of the 15th International Conference on Educational Data Mining*, pages 98–109, Durham, United Kingdom, July 2022. International Educational Data Mining Society.

[19] D. Tempelaar, B. Rienties, J. Mittelmeier, and Q. Nguyen. Student profiling in a dispositional learning analytics application using formative assessment. *Computers in Human Behavior*, 78:408–420, 2018.

[20] N. Van Halem, C. Van Klaveren, H. Drachsler, S. Schmier, and I. Cornelisz. Tracking patterns over time in student data to improve students' performance prediction. *Proceedings of the 10th International Conference on Learning Analytics and Knowledge*, pages 555–564, 2020.

[21] H. Waheed, S.-U. Hassan, N. R. Aljohani, J. Hardman, S. Alelyani, and R. Nawaz. Predicting academic performance of students from vle big data using deep learning models. *Computers in Human behavior*, 104:106189, 2020.

[22] M. Zhou and P. H. Winne. Modeling academic achievement by self-reported versus traced goal orientation. *Learning and Instruction*, 22(6):413–419, 2012.